(a) Assuming a total vehicle mass of 1500 kg, with the weight evenly distributed among four wheels, the sprung mass per wheel is approximately $m_q = 375$ kg. Assuming a static suspension compression of about $\Delta = 0.25m$. the equivalent suspension stiffness for a single wheel is

$$k_s = \frac{m_q g}{\Delta} = \frac{375 \times 9.81}{0.25} \approx 1.47 \times 10^4 \ N/m.$$

if the effective rolling radius of the tire is taken as $r_t = 0.31m$, the viheele speed as $V = 100 km/h$. then the tire rotational speed is

$$f_{rot} = \frac{v}{2\pi r_t} = \frac{27.78}{2 \times 3.14 \times 0.31} \approx 14.3 Hz. \quad w \approx 89.9 rad/s$$

(b)

Example: Quarter-car suspension model.

The quarter mass of the vehicle body can be represented by a mass block m. the suspension spring by stiffness k. the damper by viscout damping c. The free viberation ralative to the static equilibriam position is.

$$m\ddot{x} + c\dot{x} + kx = 0.$$

if $\zeta < 1$

$$x(t) = e^{-\zeta w_n t}\left[x_0 \cos w_d t + \frac{v_0 + \zeta w_n x_0}{w_d} \sin w_d t\right]$$

$$w_d = w_n \sqrt{1-\zeta^2}$$

The greater the damping, the faster the amplitude decays. excessive damping reduce wheel's ability to follow the road surface; too little damping may lead to multiple bouncing.

(1) Take vertical downward as positive and the unloaded position of the spring as zero. The absolute coordinates are satisfied

$$m\ddot{y} = mg - ky.$$

The static equilibrium position y_e satisfies $mg = ky_e$.
let $x = y - y_e$ then $dx = dy$ $ddx = ddy$. ∴

$$m\ddot{x} + kx = 0$$

∴ the constant gravitational term present in the absolute coordinate equation only alters the static equilibrium position, but doesn't affect the natural frequency $\omega_n = \sqrt{\frac{k}{m}}$.

(2).

$$\frac{1}{2} I_c \cdot \frac{L^2}{r^2} \dot{\theta}^2$$

$$V_c = L \cdot \dot{\theta} \qquad \omega_d = \frac{V_c}{r} = \frac{L}{r}\dot{\theta}$$

$$\begin{cases} T_b = \frac{1}{2} I_o \dot{\theta}^2 & I_o = \frac{1}{3} m_b L^2 \\ T_d = \frac{1}{2} m_d (\dot{\theta} L)^2 + \frac{1}{2} I_c \cdot \omega_d^2 & I_c = \frac{1}{2} m_d r^2 \end{cases}$$

$$V = L\left(\frac{m_b}{2} + m_d\right)(1 - \cos\theta)$$

$$\frac{d(T+V)}{dt} = L\left(\frac{m_b}{2} + m_d\right)\sin\theta \cdot \dot{\theta} + I_o \dot{\theta}\ddot{\theta} + \left(m_d L^2 + \frac{I_c L^2}{r^2}\right)\dot{\theta}\ddot{\theta} = 0$$

$\sin\theta \approx \theta$.

$$\therefore \quad \ddot{\theta} + \omega_n^2 \theta = 0 \qquad \omega_n^2 = \frac{g}{L} \frac{m_b/2 + m_d}{m_b/3 + 3m_d/2} \quad \approx 4.349 \text{ rad/s}.$$

$$\therefore f_n \approx 0.693 \text{ Hz}.$$

(3) $T = \frac{1}{2}\left(m_A + m_B + \frac{I}{r^2}\right)\dot{x}^2$.

$V = \frac{1}{2}k(2x)^2$ Compared to the static equilibrium position, the gravitational constant term is eliminated, thus.

$$m_{eq}\ddot{x} + 4kx = 0. \qquad m_{eq} = 7.25 \text{ kg}.$$

$$\omega_n = \sqrt{\frac{4k}{m_{eq}}} = 9.097 \text{ rad/s}. \qquad f_n = 1.448 \text{ Hz}$$

(4) $T = \frac{1}{2}mv^2 + \frac{1}{2}I\dot{\theta}^2$ $\quad v = R\dot{\theta}$.

$= \frac{3}{4}mr^2 \cdot \dot{\theta}^2$

$V = \frac{1}{2}kx^2$.

$\frac{d(V+T)}{dt} = 0 \Rightarrow \ddot{\theta} + \omega_n \theta = 0$. $\quad \omega_n = \sqrt{\frac{2k}{3m}} = 7.563\ \text{rad/s}$.

$\theta(0) = 0.2\ \text{rad} \quad \dot{\theta}(0) = 1\ \text{rad/s}$.

$\therefore \theta(t) = -0.1322\sin(7.563t) + 0.2\cos(7.563t)$

$\therefore |\dot{\theta}|_{max} = \sqrt{0.1322^2 + 0.2^2} \times 7.563 \approx 1.813\ \text{rad/s}$

(6)

$M_{eq} = m_1 + m_2 + \frac{I}{r^2} = 60 + \frac{0.03}{0.12^2} = 62.083\ kg$.

$V = \frac{1}{2}kx^2 + (m_1 g\sin 20^\circ - m_2 g)x$. $\quad T = \frac{1}{2}M_{eq}\cdot\dot{x}^2$

$x_e = \frac{m_2 g - m_1 g\sin 20^\circ}{k} = 0.2421\ m$.

$\therefore M_{eq}\cdot\ddot{q} + kq = 0 \quad (q = x - x_e)$.

$\omega_n = \sqrt{\frac{k}{m_{eq}}} = 3.590\ \text{rad/s}$. $\quad f = 0.571\ Hz$

(7)

(a) $x_k = a\theta$. $\quad x_m = b\theta$.

$I_0 = b^2 m = 1750 \times 2.5^2 = 10937.5\ kg\cdot m^2$.

$F_c = c\cdot b\cdot\dot{\theta} \quad \therefore \quad M_c = F_c\cdot b = cb^2\dot{\theta} \quad \therefore C_\theta = cb^2$

$F_k = ka\theta \quad M_k = F_k\cdot a = ka^2\theta$. $\quad \therefore K_\theta = ka^2$.

$\therefore mb^2\ddot{\theta} + cb^2\dot{\theta} + ka^2\theta = 0$.

$\therefore \ddot{\theta} + 2\dot{\theta} + 100\theta = 0$

(b) $\omega_n = \sqrt{100} = 10\ \text{rad/s}$. $\quad \xi = \frac{cb^2}{mb^2}/(2\omega_n) = 0.1 < 1$.

$\therefore$ Underdamped system.

$\omega_d = \omega_n\sqrt{1-\xi^2} = 9.95\ \text{rad/s}$

$f_d = 1.58\ Hz$

$\theta(t) = e^{-\xi\omega_n t}(A\cos\omega_d t + B\sin\omega_d t) \quad \theta(0) = 0.1 \quad \dot{\theta}(0) = 0$.

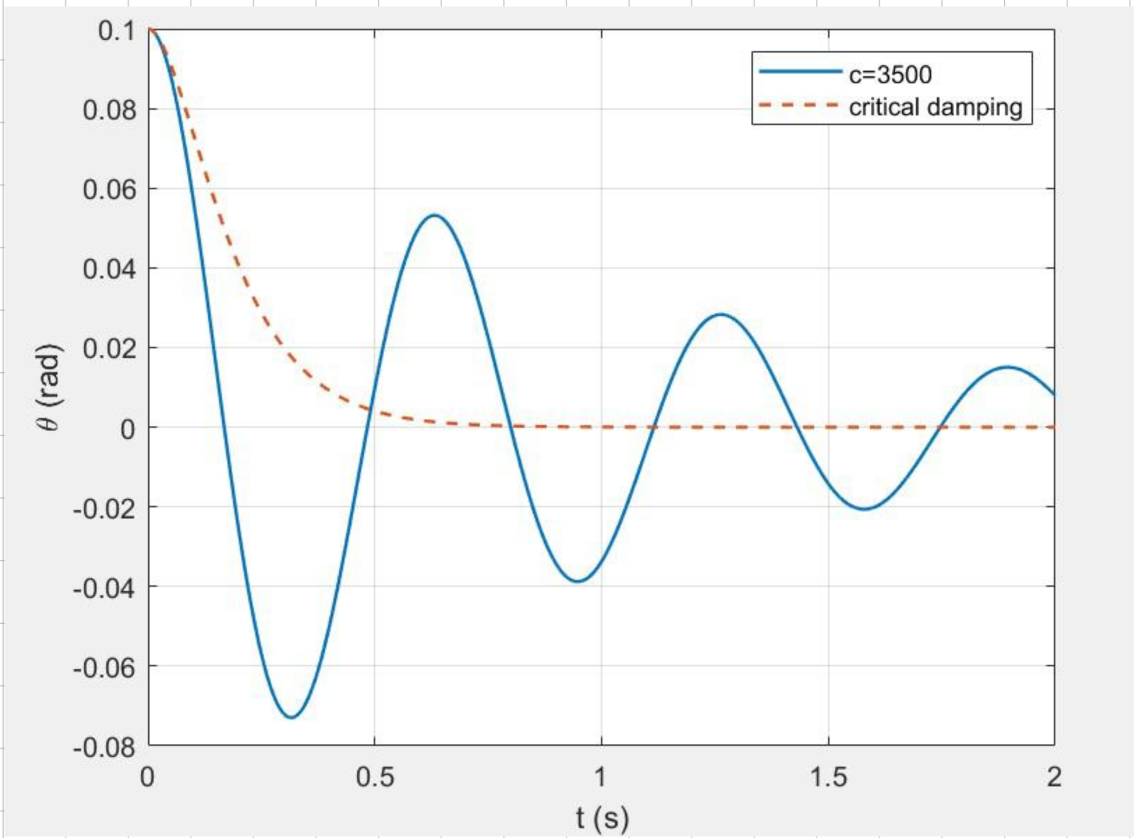
$\therefore A = 0.1 \quad B = \frac{\xi\omega_n A}{\omega_d} = 0.01005$

$\therefore\ \theta(t) = 0.1e^{-t}\left[\cos(9.95t) + 0.1005\sin(9.95t)\right]$ rad

(c) $C_{\theta,cr} = 2\sqrt{I_0 k_0} = 218750\ m \cdot N \cdot s/rad,$

$C_{cr} = C_{\theta cr}/b^2 = 35000\ N \cdot s/m.$

$\frac{c}{C_{cr}} = 0.1 = \zeta$ $\quad\therefore$ is Underdamped system.



(0)

(a) Example:

Hydraulic shock absorbers in vehicles

Hydraulic door closers

Dashpots used in mechanical vibration isolators.

Hydraulic vibration dampers in buildings and bridges.

(b)

Material	ζ
steel	0.1% – 0.5%
Aluminum	0.1% – 0.3%
Cast iron	0.5% – 2%
conrete	2% – 7%
wood	1% – 5%

(1)

(a) $T = \frac{1}{2} I \dot{\theta}^2 \quad V = \frac{1}{2} k (2R\theta)^2 = 2kR^2\theta^2$

Rayleigh dissipative function: $D = \frac{1}{2} c \cdot (R\dot{\theta})^2$

Lagrange function: $\frac{d}{dt}\frac{\partial T}{\partial \dot{\theta}} - \frac{\partial T}{\partial \theta} + \frac{\partial V}{\partial \theta} + \frac{\partial D}{\partial \dot{\theta}} = 0$

$\therefore \; I\ddot{\theta} + cR^2\dot{\theta} + 4kR^2\theta = 0$

(b) $k_\theta = 4kR^2 \quad C_\theta = cR^2 \quad \therefore C_{cr} = \frac{2\sqrt{I k_\theta}}{R^2} = 277.1 \text{ N}\cdot\text{s/m}$

$\omega_n = \sqrt{\frac{4kR^2}{I}} = 4.330 \text{ rad/s}$

(c) $\theta(0) = 0 \quad \dot{\theta}(0) = 10$

$\therefore \theta(t) = 10t\,e^{-4.330t}$ rad

(d) $\dot{\theta} = 10 e^{-\omega_n t}(1-\omega_n t)$. $t_{max} = 1/\omega_n = 0.2309\ s$

$\theta_{max} = \frac{10}{\omega_n e} = 0.8509\ rad$.

(e) $e \cdot \omega_n t \cdot e^{-\omega_n t} = 0.1$ $\omega_n t = 4.8897$. $\therefore t_{10\%} = 1.129\ s$.

(f) Let $k' = 300\ N/m$. $\Rightarrow \omega_n' = 6.124\ rad/s$. $\zeta' = 0.7071$ $\omega_d' = 4.330\ rad/s$.

$\theta(t) = \frac{10}{4.330} e^{-4.330t} \sin(4.330t)$

$e \cdot \omega_n t \cdot e^{-\omega_n t} = 0.1 \Rightarrow t'_{10\%} = 0.615\ s$

(2)

$m\ddot{x} + c\dot{x} + kx = mg$. $\omega_n = \sqrt{\frac{k}{m}} = 24.721\ rad/s$ $\zeta = \frac{c}{2m\omega_n} = 1.236 > 1$

$\therefore$ is overdamping system.

$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1} = -12.5964, -48.5147\ s^{-1}$

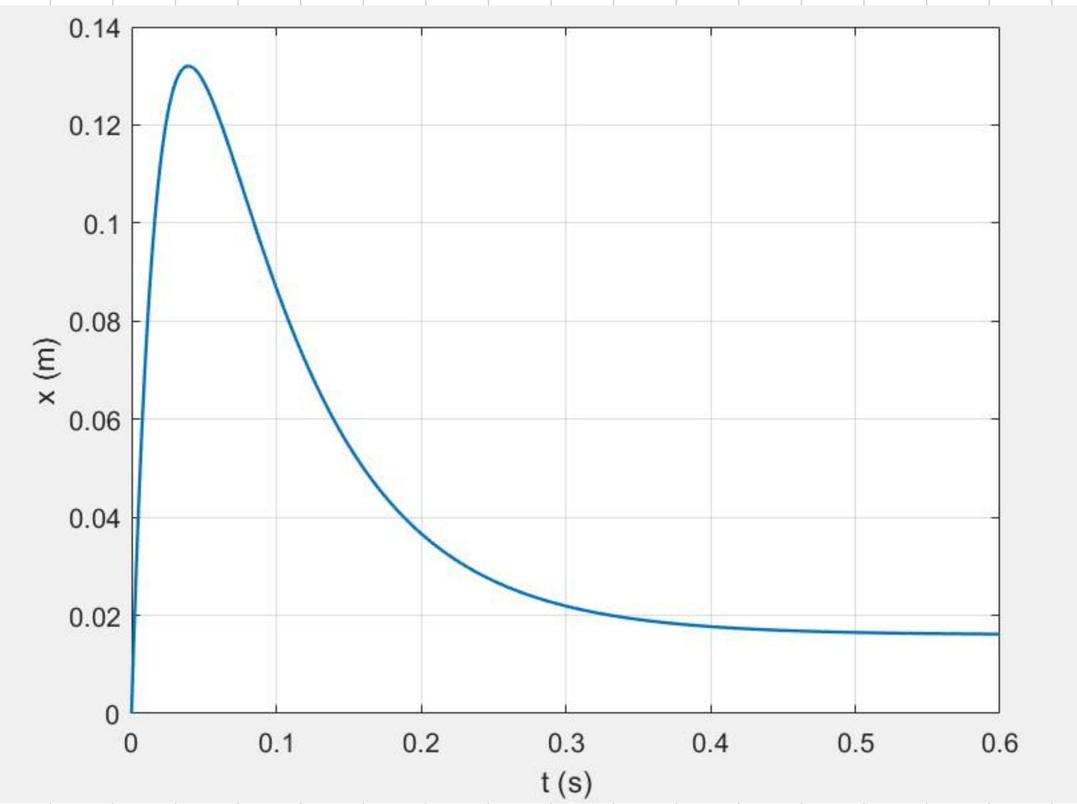
Let $q = x - x_e$ $q(0) = -x_e$ $\dot{q}(0) = 10$.

$\therefore$ $q(t) = 0.2567 e^{-12.5964t} - 0.2778 e^{-48.5147t}\ m$

$\dot{q}(t) = 0 \Rightarrow t_{max} \approx 0.0393\ s$ $q_{max} \approx 0.116\ m$

$0.1\, q_{max} = 0.0116\ m$. $\therefore q(t_{10\%}) = 0.0116\ m$.

$t_{10\%} \approx 0.246\ s$.



(3) $f = \frac{6000}{60} Hz = 100 Hz$

$z = x - y \qquad \therefore x = z + y \qquad \dot{x} = \dot{z} + \dot{y}$

$\ddot{x} = \ddot{z} + \ddot{y}$

$m\ddot{x} = -kz - c\dot{z}$

$\therefore m\ddot{z} + c\dot{z} + kz = -m\ddot{y} \qquad Z \to i\omega Z \to -\omega^2 Z$

$-m\omega^2 Z + i\omega c Z + kZ = m\omega^2 Y$

$\therefore |Z| = \frac{m\omega^2 Y}{\sqrt{(k - m\omega^2)^2 + \omega^2 c^2}}$

$y(t) = Y \sin n\omega t$

$kz \approx -m\ddot{y} \qquad \therefore |\ddot{y}| \approx \frac{k}{m} z = \omega_n^2 z$

$\frac{a_{ind}}{a} = \frac{\omega_n^2 Z}{\omega^2 Y} = \frac{m}{\sqrt{(k - m\omega^2)^2 + \omega^2 c^2}} \qquad \because k = m\omega_n^2,\ r = \frac{\omega}{\omega_n},\ \zeta = \frac{c}{2m\omega_n}$

$\therefore = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}$

$= \frac{10}{9.81}$

$\because \omega_d = \omega \qquad \omega_d = \omega_n \sqrt{1 - \zeta^2} \qquad r = \frac{\omega}{\omega_n} = \frac{\omega_d}{\omega_n}$

$\therefore \Rightarrow 4\zeta^2 - 3\zeta^4 = 0.981^2$

$\zeta \approx 0.561 \qquad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} \approx 759.2 \text{ rad/s}$

$f_n \approx 120.8 Hz \qquad k = m\omega_n^2 \approx 5.76 \times 10^3 N/m$

$c = 2\zeta m\omega_n = 8.52 Ns/m$

(4) $r_A\theta_A + r_B\theta_B = 0$. $\therefore \theta_B = -\frac{r_A}{r_B}\theta_A$. $\dot{\theta}_B = -0.7\dot{\theta}_A$

$T_A = \frac{1}{2}I_A\dot{\theta}_A^2$ $T_B = \frac{1}{2}I_B\dot{\theta}_B^2$ $I_{eq} = I_A + 0.7^2 I_B = 0.074\,kg\cdot m^2$

$P_d = C_B\dot{\theta}_B^2 = C_B(0.7\dot{\theta}_A)^2$ $\therefore C_\theta = 2\cdot 0.7^2 = 0.98\,Nms/rad$

$\therefore I_{eq}\cdot\ddot{\theta}_A + C_\theta\dot{\theta}_A + k\theta_A = 0$

$\therefore 0.074\ddot{\theta}_A + 0.98\dot{\theta}_A + 10\theta_A = 0$

$\therefore \omega_n = \sqrt{\frac{10}{0.074}} = 11.62\ rad/s$ $\xi = \frac{C_\theta}{2\sqrt{kI_{eq}}} = \frac{0.98}{2\times\sqrt{10\times 0.074}} \approx 0.57$ $\begin{cases}0.\\<1\end{cases}$

$\omega_d = \omega_n\sqrt{1-\xi^2} \approx 9.55\ rad/s$ $f_d = 1.52Hz$

(5) $\therefore m_1v_1^- + m_2v_2^- = (m_1+m_2)v_0$. $v_1^- = 0$ $v_2^- = \sqrt{2gh}$.

$\therefore v_0 = \frac{m_2}{m_1+m_2}\sqrt{2gh}$.

$M = m_1 + m_2$

$M\ddot{x} + c\dot{x} + kx = Mg$. $kx_e = Mg$

$q(t) = x(t) - x_e$ $\therefore M\ddot{q} + c\dot{q} + kq = 0$.

$x(0) = \frac{m_1 g}{k}$ $\therefore q(0) = \frac{m_1 g}{k} - \frac{(m_1+m_2)g}{k} = -\frac{m_2 g}{k}$.

$\dot{q}(0) = v_0 = \frac{m_2}{m_1+m_2}\sqrt{2gh}$.

① if $\xi < 1$. $\omega_n = \sqrt{\frac{k}{m_1+m_2}}$ $\xi = \frac{c}{2\sqrt{kM}} = \frac{c}{2\sqrt{k(m_1+m_2)}}$

$\omega_d = \omega_n\sqrt{1-\xi^2}$

$q(t) = e^{-\xi\omega_n t}[A\cos\omega_d t + B\sin\omega_d t]$ $\begin{cases} q(0) = -\frac{m_1 g}{k}. \\ \dot{q}(0) = v_0 \end{cases}$

$$\Rightarrow \quad B = \frac{\frac{m_2}{m_1+m_2}\sqrt{2gh} - \xi\omega_n\frac{m_2 g}{K}}{\omega_d}$$

$$\therefore q(t) = e^{-\xi\omega_n t}\left[-\frac{m_2 g}{K}\cos(\omega_d t) + \frac{\frac{m_2}{m_1+m_2}\sqrt{2gh} - \xi\omega_n\frac{m_2 g}{K}}{\omega_d}\sin(\omega_d t)\right]$$

$$\left(\xi = \frac{c}{2\sqrt{k(m_1+m_2)}} \qquad \omega_n = \sqrt{\frac{K}{m_1+m_2}} \qquad \omega_d = \omega_n\sqrt{1-\xi^2}\right)$$

② if $\xi > 1$

$$q(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}. \qquad \lambda_{1,2} = -\omega_n(\xi \pm \sqrt{\xi^2-1}).$$

$$C_1 = \frac{v_0 + \lambda_2\frac{m_2 g}{K}}{\lambda_1 - \lambda_2} \qquad C_2 = \frac{-m_2 g}{K} - C_1$$

(b)

(a) $T \approx 0.1\,S.$

$$\delta = \frac{1}{N}\ln\left(\frac{x_i}{x_{i+N}}\right) = \frac{1}{3}\ln\left(\frac{16}{6}\right) \approx 0.33$$

$$\delta = \frac{2\pi\xi}{\sqrt{1-\xi^2}} \qquad \therefore \xi \approx 0.052 << 1$$

$$\omega_n = \cancel{\omega_d}\sqrt{1-\xi^2} = 62.8/\sqrt{1-0.052^2} \approx 62.9\,\text{rad/s}.$$

$$f_n \approx 10.0\,\text{Hz}$$

Envelop equation:

b) $$q(t) = Ae^{-\xi\omega_n t} \qquad A \approx 20\,\text{mm}$$

$$20e^{-3.3t} = 0.01 \qquad \therefore t \approx \frac{1}{3.3}\ln\left(\frac{20}{0.01}\right) \approx 2.30\,S$$

(c) $k' = 2k \quad m' = \frac{1}{2}m \quad c' = c$

$\therefore \quad \omega_n' = 2\omega_n \quad \xi' = \xi \quad \Rightarrow \therefore \omega_d' = 2\omega_d.$

$\therefore \quad q(t) = 20e^{-6.6t} = 0.1. \quad t' = \frac{1}{2}t \approx 1.15s.$

(d) $q(t) = e^{-\xi\omega_n t}\left[q_0 \cos(\omega_d t) + \frac{v_0 + \xi\omega_n q_0}{\omega_d}\sin(\omega_d t)\right]$

$q_0 = -10mm$

$\dot{q}_0 = -\xi\omega_n A + \omega_d B$

when $t = 0.06s \quad q(t) \approx 16mm.$

$16 = e^{-3.3\times 0.06}\left[-10\cos(62.8\times 0.06) + B\sin(62.8\times 0.06)\right]$

$\Rightarrow B \approx -19.5mm.$

$\therefore \dot{q}_0 = V_0 \approx -3.3\times(-10) - 19.5\times 62.8 \approx -1.2 m/s.$